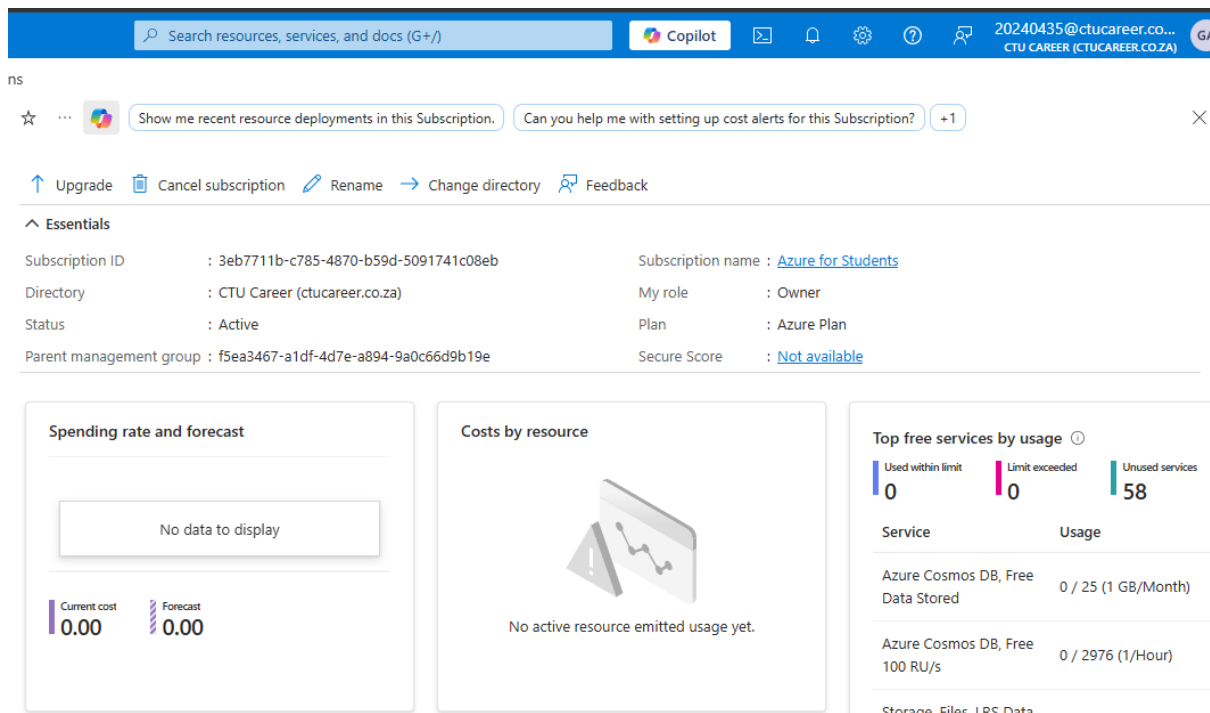


Week 1



2.

CTU Training Solutions

1. Type of Organisation

CTU Training Solutions is a private education organisation in South Africa. They teach students from places. The organisation wants to give students education and practical skills. They use technology a lot to help with teaching and managing students.

CTU has students and staff members in different locations. So they need an IT system that works well and can handle a lot of users.

2. Main IT Needs of the Organisation

CTU Training Solutions needs an IT system for teaching and administration. Their main IT needs are:

Secure storage of student and staff data.

Reliable access to online learning systems.

Communication platforms for lecturers, students and staff.

Virtual machines for IT training.

Backup solutions to protect data.

A good network that supports campuses.

Identity and access management to keep systems secure.

Monitoring tools to detect threats.

The organisation must support a growing number of users while keeping systems stable.

3. Why Azure is Better than On-Premises Infrastructure

Microsoft Azure is better for CTU Training Solutions than infrastructure.

Azure is scalable so CTU can. Decrease resources as needed. This helps during periods like student registration.

Azure reduces infrastructure costs because CTU does not need to maintain physical servers.

Azure improves accessibility because staff and students can access resources from anywhere.

Azure provides built-in backup, disaster recovery, monitoring and security features.

4. Security and Compliance Requirements

CTU Training Solutions must protect information like student records and financial data.

Important security requirements include:

- factor authentication for user accounts.

- Role-based access control to limit user permissions.

- Encryption of data.

- Firewall protection.

- Regular. Disaster recovery planning.

- Monitoring and threat detection.

- Compliance with African data protection regulations.

These measures help keep data safe.

5. Expected Number of Users

The expected number of users accessing the cloud environment includes like

- Students

- Lecturers

- Staff

IT support staff

Management personnel

CTU may have several thousand users accessing services. A reasonable estimate would be, between 2,000 and 10,000 users. The Azure environment must be scalable and capable of supporting users while maintaining performance and security.

3.

Azure Service	Purpose / Why you need it	Week it is used
Resource Group	A Resource Group is used to organise and manage all Azure resources related to the project in one location. It helps with administration, monitoring, and access control.	1
Virtual Network (VNet)	A Virtual Network is really important because it helps to keep things safe when Azure resources like machines and storage services talk to each other. It is, like building the projects private network on the cloud.	2
Virtual Machine (VM – B1s)	The B1s Virtual Machine is an affordable way to get a cloud server. You can use the B1s Machine for trying out new things, managing systems and doing hands on work. The B1s Virtual	3

	Machine can run Windows Server or Ubuntu Linux.	
Microsoft Entra ID	Microsoft Entra ID is a tool that helps with identity and access management for Microsoft Entra ID users. It does a few things, like authentication and permissions to make sure the sign-in process is secure when you use Azure services, with Microsoft Entra ID.	4
Network Security Group (NSG)	Network Security Groups are used to control the traffic that comes into and goes out of a network, through security rules. This makes the network safer. Stops people who are not allowed from getting in. Network Security Groups do a job of protecting the network.	2 and 3
Azure Storage Account	Azure Storage Accounts provide secure cloud storage for files, backups, logs, and virtual machine data	5
Microsoft Defender for Cloud	Microsoft Defender for Cloud helps keep an eye on security risks. It gives recommendations to make things better. This tool improves protection for Azure resources.	6

	Microsoft Defender, for Cloud is really helpful	
Azure Monitor	Azure Monitor is used to collect logs, performance metrics, and alerts to monitor the health and performance of Azure resources.	7
Azure Cost Management	Azure Cost Management is a tool that helps people keep an eye on how much they're spending on the cloud. It also helps them monitor their budgets and cut out any costs that they do not need. This is all done within the Azure subscription. Azure Cost Management is very useful, for managing Azure costs.	8

4.

Search resources, services, and docs (G+/)
Copilot
20240435@ctucareer.co...
CTU CAREER (CTUCAREER.CO.ZA)

Resource groups

Summarize my costs by service
Export resource groups using Bicep or Terraform
How to manage changes with deployment tools?

+ Create
Manage view
Refresh
Export to CSV
Open query
Assign tags
Add to service group
Group by none

Filter for any field...
Subscription equals all
Location equals all
Add filter

☐ Name ↑	Subscription	Location
☐  CTU-20240435-RG	... Azure for Students	South Africa North

Name ⓘ	Value ⓘ
Module	: NET731
Project	: NET731-Capstone
Student	: Godfrey Anderson